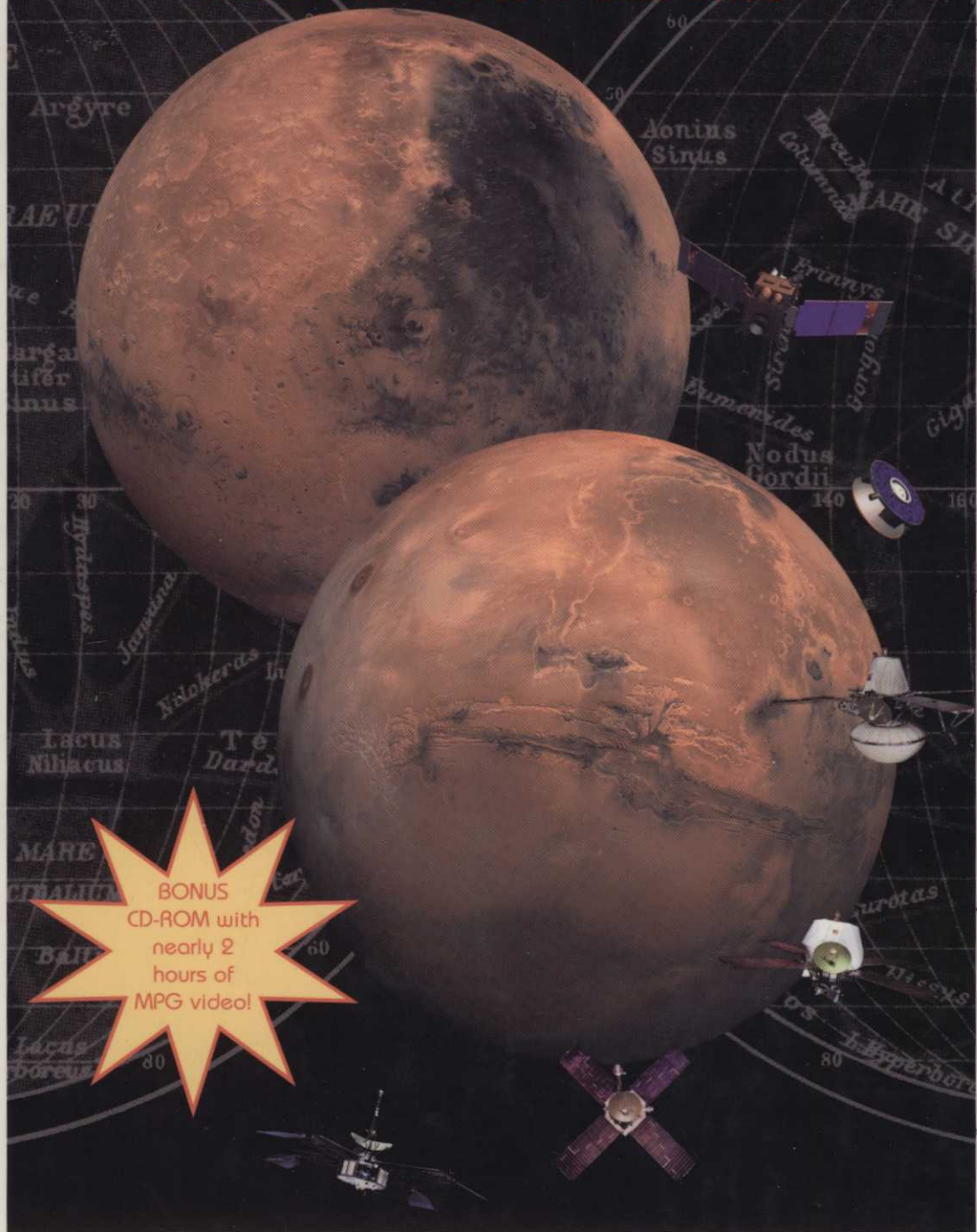


MARS

THE NASA MISSION REPORTS



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hours of
MPG video!

Pathfinder Mission Status Tuesday, March 10, 1998

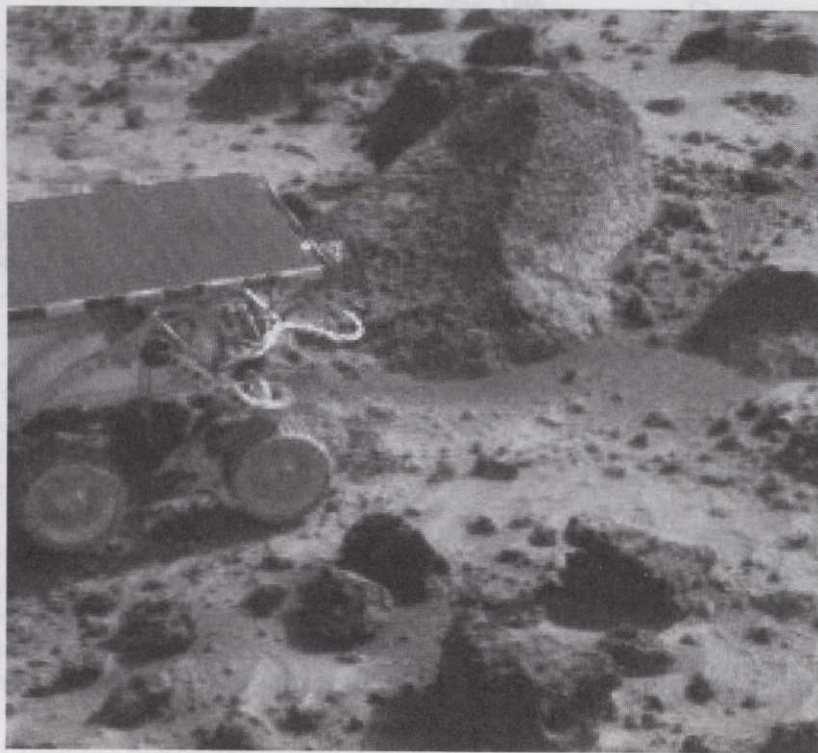
ing goodbye to NASA's Mars Pathfinder lander and the Sojourner rover ended today when the lander
to respond to the final command to communicate with controllers at NASA's Jet Propulsion
tory. The Pathfinder mission, which operated three times longer than its original 30-day planned
on the Martian surface, is acknowledged as one of NASA's most successful endeavors as a dramatic
e of the space agency's new style of "faster, better, cheaper" planetary exploration.

last-ditch effort to listen for a signal from Pathfinder effectively ends the mission, said project manager
Muirhead. No further attempts will be made to communicate with Pathfinder, he added.

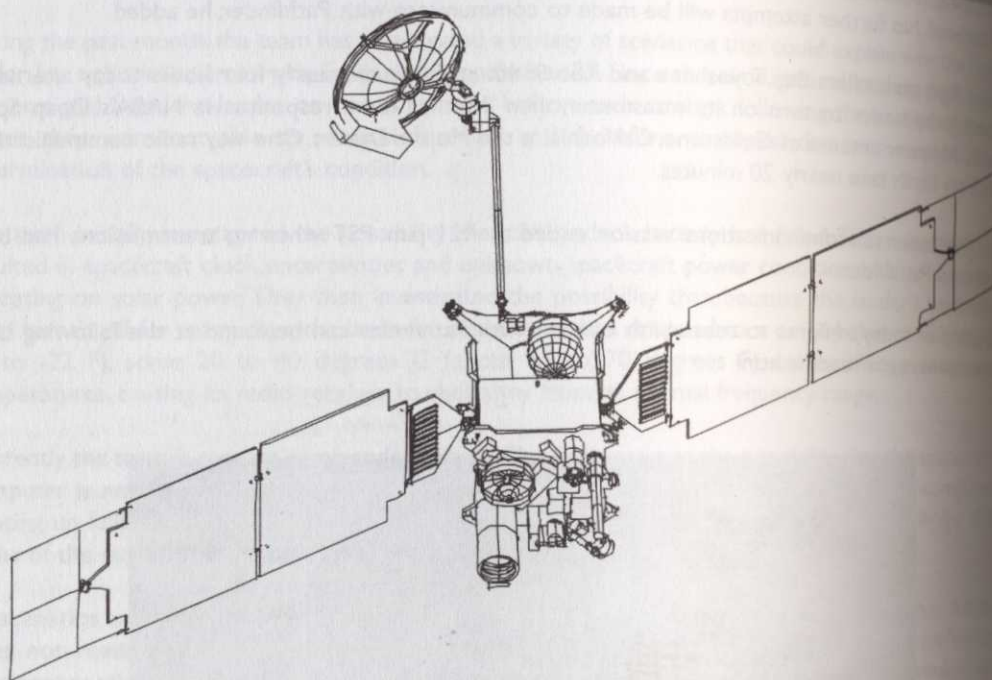
nder flight controllers Ben Toyoshima and Rob Smith at JPL spent nearly four hours today alternately
anding the lander to turn on its transmitter, then listening for a response via NASA's Deep Space
ork's 34-meter antenna at Goldstone, California, in the Mojave Desert. One-way radio communications
rs from Earth take nearly 20 minutes.

nal Pathfinder telecommunications session ended at 1:21 p.m. PST when no transmissions had been
ced from Pathfinder.

cription of today's efforts to reestablish contact with Pathfinder can be found at the following URL:
mars.jpl.nasa.gov/readme.html



Mars Global Surveyor Arrival Press Kit



NATIONAL AERONAUTICS AND SPACE ADMINISTRATION

RELEASE: September 1997

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MARS GLOBAL SURVEYOR ON TARGET FOR ARRIVAL

NASA's Mars Global Surveyor, the first in a new series of spacecraft destined to explore the red planet, is preparing to intercept the orbit of Mars and begin a two-year mapping mission after a 10-month, 435-million-kilometer (435-million-mile) interplanetary journey.

The orbiter will fire its main engine beginning at 01:17 Universal Time on September 12 (6:17 p.m. Pacific Daylight Time September 11) for 22 minutes to slow its speed enough to be captured in orbit around the planet.

Mars Global Surveyor is a global mapping mission, carrying a suite of science instruments designed to study the entire Martian surface, atmosphere and interior. Measurements will be collected beginning in March 1998 from a low-altitude, nearly polar orbit 378 kilometers (234 miles) above the Martian surface over the course of one complete Martian year, the equivalent of nearly two Earth years.

"Throughout its primary, two-year mission, Mars Global Surveyor will gather information on the geology, geophysics and climate of Mars," said Glenn E. Cunningham, Global Surveyor project manager at NASA's Jet Propulsion Laboratory, Pasadena, CA.

"The mission will provide a global portrait of Mars as it exists today," he said. "This new view will help planetary scientists to better understand the history of Mars' evolution, and will provide clues about the planet's interior and surface evolution. With this information, we will have a better understanding of the history of all of the inner planets of the solar system, including our home planet, Earth."

Mars Global Surveyor continues NASA's long exploration of the red planet, which began more than 30 years ago with the Mariner 4 spacecraft that produced the first pictures of the planet's cratered surface. Following the successful landing of the Mars Pathfinder lander and rover on July 4, 1997, Mars Global Surveyor is the first in a multi-year series of missions called the Mars Surveyor program that will lead to eventual human expeditions to the red planet.

Mars Global Surveyor was launched at 12:00:49 p.m. Eastern Standard Time on November 7, 1996, atop a three-stage Delta II launch vehicle from launch pad 17A at Cape Canaveral Air Station, FL. The third-stage Star 48B solid rocket later propelled the spacecraft out of Earth orbit and on its way to Mars.

Once on course for the cruise to Mars, the spacecraft deployed its two solar panels to begin generating solar power. One of the solar panels did not fully deploy and is tilted about 20 degrees from its intended position; this is not, however, expected to pose a significant risk to the mission. The low-gain antenna was used for initial spacecraft communications, until the spacecraft was far enough away from Earth in early January 1997 to begin using its 1.5-meter-diameter (5-foot) high-gain antenna.

Mars Global Surveyor's six science instruments — the thermal emission spectrometer, laser altimeter, magnetometer/electron reflectometer, ultra-stable oscillator, camera and radio relay system — were calibrated during the cruise to Mars. Three trajectory correction maneuvers were performed to fine-tune the spacecraft's flight path. All spacecraft systems and the instrument payload performed well as Mars Global Surveyor headed for its destination, according to Joe Beerer, Global Surveyor flight operations manager.

When Surveyor reaches Mars, its 660-newton main engine will fire to slow the spacecraft's speed by more than 973 meters per second (2,176 miles per hour) with respect to Mars and allow the craft to be captured in Mars' gravity.

"Mars Global Surveyor will be flying over the north pole when it enters orbit around Mars," said Wayne Lee, Mars Global Surveyor mission planner. "The spacecraft will spend the first six days in this highly elliptical orbit around the planet, completing one orbit around Mars in about 45 hours, or just less than two days."

Instrument calibrations and some science measurements will take place during the elliptical orbit phase, said

day-afternoon. This "sun-synchronous" orbit puts the Sun at a standard angle above the horizon in each image.

Experiment teams will control their spaceborne instruments from their home institutions. Each 24 hours worth of data will be transmitted to Earth during daily, 10-hour tracking passes performed by NASA's Deep Space Network.

The Mars Global Surveyor mission is expected to yield more than 700 billion bits of scientific data, more than the amount of data returned by all previous Mars missions, and exceeded only by the Magellan Venus mission.

Mars Global Surveyor will examine the entire planet — from the ionosphere, an envelope of charged particles surrounding Mars, down through the atmosphere to the surface and deep into Mars' interior. Scientists will glean valuable new information on daily and seasonal weather patterns, geological features and the migration of water vapor from hemisphere to hemisphere over a complete Martian year.

As the primary mission winds down in late 1999, Global Surveyor will be used to relay data from microprobes penetrators beneath the surface of Mars that will have been deployed before arrival by the 1998 Mars Surveyor, and will be used as a backup relay platform for data from the Mars Surveyor '98 lander itself. Depending on its lifetime, Global Surveyor may also serve as a communications relay station for other spacecraft arriving at Mars.

Mars Global Surveyor is the first mission in a sustained program of robotic exploration of Mars, managed by the Jet Propulsion Laboratory, Pasadena, CA, for NASA's Office of Space Science, Washington, DC. JPL is a division of the California Institute of Technology.

[End of General Release]

Media Services Information

NASA Television Transmission

NASA Television is broadcast on the satellite GE-2, transponder 9C, C Band, 85 degrees west longitude, frequency 3880.0 MHz, vertical polarization, audio monaural at 6.8 MHz. Television coverage of the Mars Global Surveyor orbit insertion begins at 5 p.m. Pacific Daylight Time on Sept. 11 and concludes at approximately 8:30 p.m. PDT, following a live news briefing. The schedule for NASA TV programming is available from the Jet Propulsion Laboratory, Pasadena, CA; Johnson Space Center, Houston, TX; Kennedy Space Center, FL; and NASA Headquarters, Washington, DC.

Status Reports

Status reports on Mars Global Surveyor will be issued by the Jet Propulsion Laboratory's Public Information Office. They may be accessed online as noted below. Daily audio status reports are available by calling (800) 665-6654.

Status Briefings

The pre-arrival mission and science overview briefing, originating from the Jet Propulsion Laboratory, will be held live on NASA TV at 10 a.m. Pacific Daylight Time on Tuesday, September 9. Status briefings will be held at 10 a.m. PDT Wednesday and Thursday, September 10-11. A post-arrival briefing will be held at 7:40 p.m. PDT Thursday, September 11. A briefing presenting initial science findings is tentatively scheduled the week of September 22-26.

Image Releases

Images returned by Mars Global Surveyor will be released to the news media in electronic format only during

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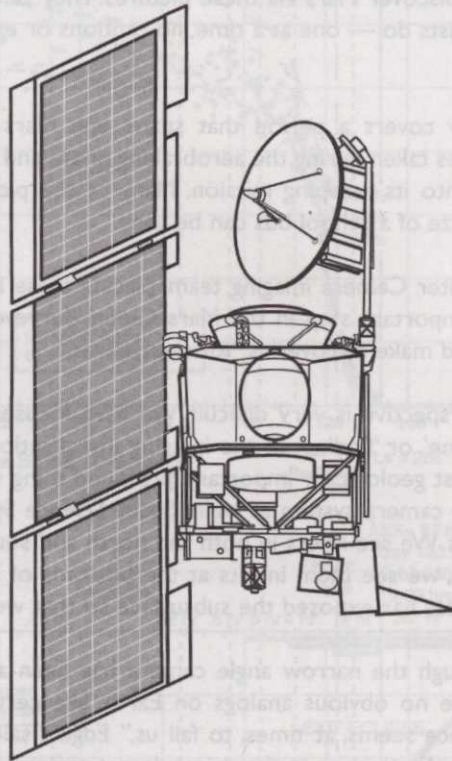
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Mars Climate Orbiter Arrival Press Kit



NATIONAL AERONAUTICS AND SPACE ADMINISTRATION

RELEASE: September 1999

Contacts

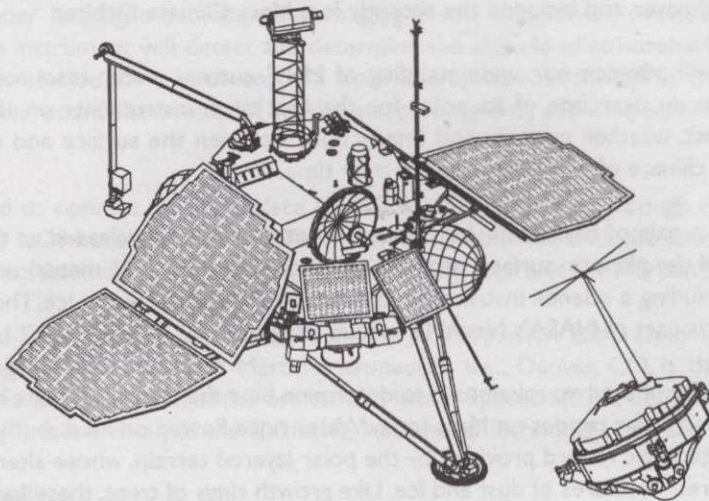
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Mars Polar Lander/ Deep Space 2 Press Kit

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION



RELEASE: December 1999

Contacts

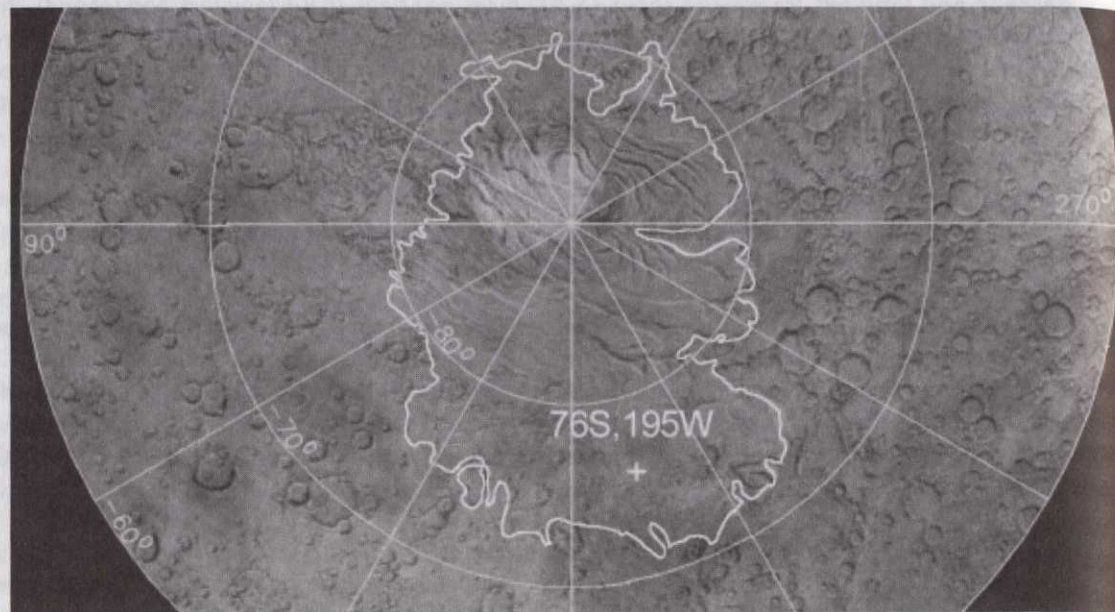
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Lander's target landing site

Shortly after landing, software which puts the spacecraft in safe mode in reaction to unexpected events will be reenabled. The descent imager will be turned off 60 seconds after landing.

After waiting five minutes to allow for dust kicked up by the landing to settle, the lander's solar arrays will be unfolded. Eight minutes after landing, while the medium-gain antenna is being turned to point at Earth, the spacecraft's gyros will be used like compasses to determine which way is north. The spacecraft's inertial measurement units will then be powered off.

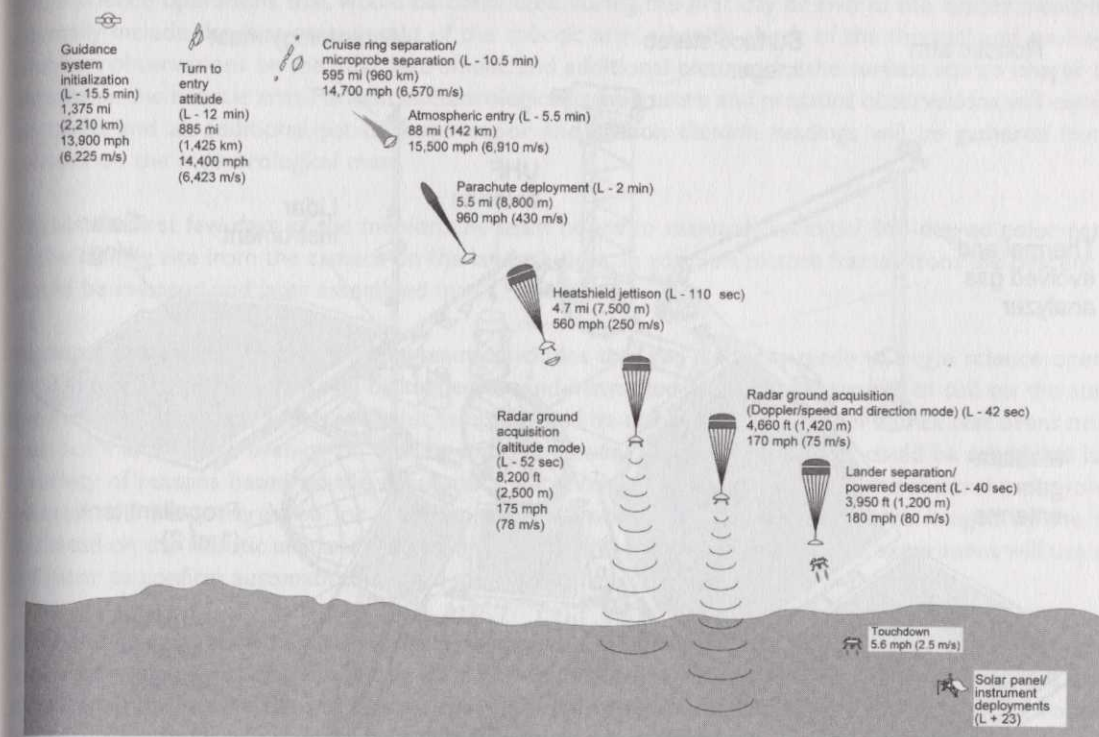
After gyrocompassing is completed, the medium-gain antenna will turn to point to Earth. This antenna slew may take up to 16 minutes to complete. A vertical scan will then be taken by the surface stereo imager before its boom is deployed. Both the meteorological and imager masts will then be raised.

First signal. The first opportunity to hear from the lander will take place when it begins transmitting directly to Earth using its dish-shaped medium-gain antenna about 23 minutes after landing; this signal is expected to be received at 12:39 p.m. PST. This transmission session will end 45 minutes later, or 1:24 p.m. PST, and would include engineering data on the lander's entry, descent and landing, as well as a possible low-resolution black-and-white picture from the undeployed camera on the lander's deck. At 1:46 p.m. PST, the lander shuts down and "sleeps" for four hours and 40 minutes while its solar panels recharge its onboard battery. No wakeups occur during this time.

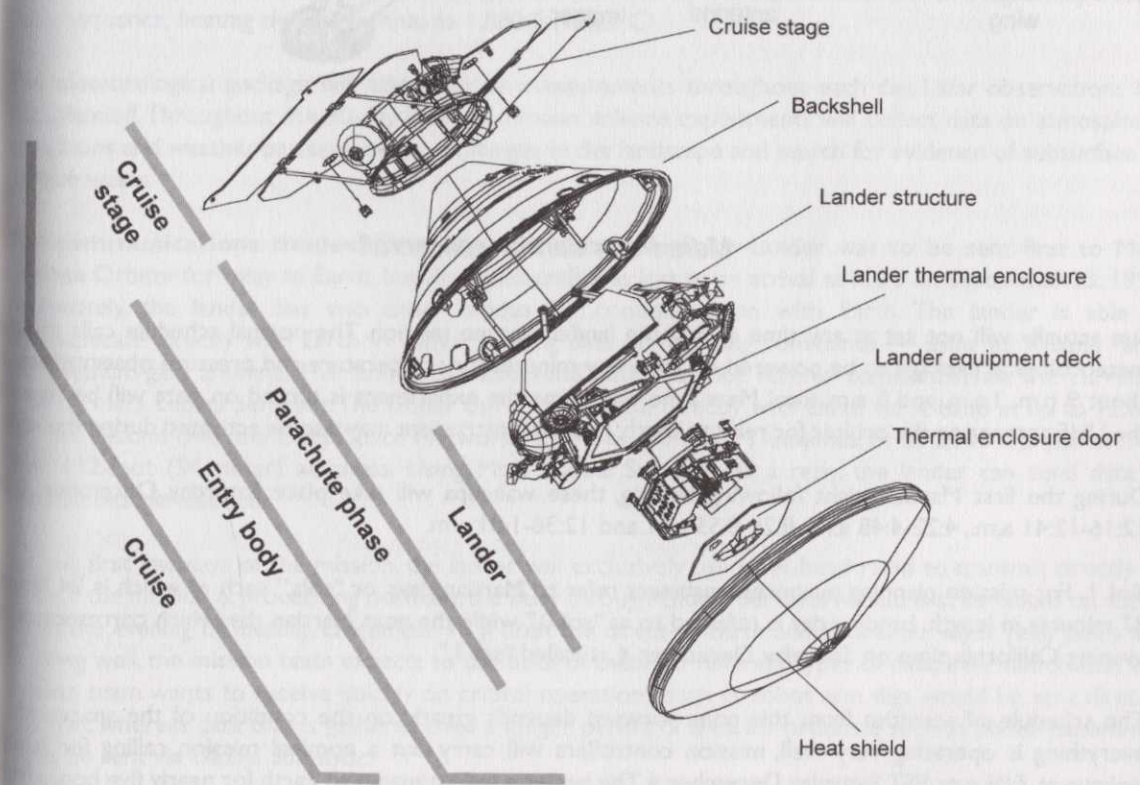
Assuming that all is normal with the spacecraft, it will power up again at 6:26 p.m. PST and turn on its receiver. At this time, mission controllers expect that they would send the lander commands such as what data rate to use for later radio transmissions. The receiver will continue listening for commands from Earth until 7:41 p.m. PST.

At 8:09 p.m., the lander will begin transmitting to Earth until 10:45 p.m.. After that session concludes, the lander will run through a sequence that takes about half an hour as it prepares to shut down for the night. At 11:24 p.m. PST the lander will power down.

Nighttime activities. The lander's onboard computer and meteorological package will be activated several times during the Martian night. Because of the extreme latitude during the Martian southern summer, the



Entry, descent and landing



Lander flight system

MANNED MARS LANDING

PRESENTATION TO THE SPACE TASK GROUP

BY DR. WERNHER von BRAUN

AUGUST 4, 1969

(Color pictures associated with this document are at the end of this book. Picture numbers are in square brackets after each title.)

INTEGRATED PROGRAM

With the recent accomplishment of the manned lunar landing, the next frontier is manned exploration of the planets. Perhaps the most significant scientific question is the possibility of extraterrestrial life in our solar system. Manned planetary flight provides the opportunity to resolve this universal question thus capturing international interest and cooperation. The information presented here describes a method of landing man on the planet Mars in 1982. The scientific goals of the mission are described and the key decision dates are identified. The unmanned planetary missions described in Part I are critical to the final designs selected.

The 1981 manned Mars mission (1982 landing on Mars) is shown as an integral part of the total space program for the next two decades. The systems and experience resulting from the Apollo program and the missions proposed for the 1970's provide the technical and programmatic foundation for this undertaking. A 1982 manned Mars landing is a logical focus for the programs of the next decade. Although the undertaking of this mission will be a great national challenge, it represents no greater challenge than the commitment made in 1961 to land a man on the moon.

MANNED PLANETARY MISSION - 1981 MARS LANDING MISSION PROFILE [1]

Several different modes are possible for accomplishing Mars landing missions, each with its peculiar advantages and disadvantages. The typical Mars Landing Mission begins with the boost of the planetary vehicle elements into Earth orbit (1) utilizing the Saturn V and Space Shuttle vehicles. Following assembly of the complete planetary vehicle in Earth orbit (2), the Earth departure phase of the mission is initiated (3). The Mars vehicle then begins a 270-day journey to Mars. This is by no means an idle phase of the mission. In addition to observations of Mars, many other experiments and measurements will be made on both the Earth-to-Mars and Mars-to-Earth legs of the trip that are of prime scientific importance. The spacecraft represents a manned laboratory in space, free of the disturbing influences of the Earth. The fact that there will be two observation points, Earth and the spacecraft, permits several possible experiments regarding the temporal and spatial features of the interplanetary environment. In addition, the spacecraft can be used to supplement and extend numerous observations conducted from Earth orbital space stations, particularly in the field of astronomy. It is possible, for example, that as yet unidentified comets might be observed for the first time.

Upon arriving at Mars, the space ship is propulsively braked into orbit (4) in the same fashion that the Apollo moon ship is placed into lunar orbit. The ship remains in Mars orbit for about 80 days (5), during which time the Mars Surface Sample Return (MSSR) probes and the Mars Excursion Modules (MEM) are deployed and the surface exploration takes place. At the end of the Mars capture period, the spacecraft is boosted out of Mars orbit (6). The return leg of the trip lasts about 290 days, during which many experiments and observations are again conducted. A unique feature of the homeward trip is a close encounter with Venus (7), about 120 days after departing Mars. Probes will be deployed at Venus during this passage, in addition to the radar mapping measurements that will be made.

The approximate two-year journey ends with the return to Earth orbit (8), and following medical examinations the crew will be returned to Earth via a Space Shuttle.

EARTH ORBIT DEPARTURE [2]

The Mars Landing Mission can be accomplished with a single planetary vehicle assembled in Earth orbit. There

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are, however, advantages in deploying two ships on the mission because of the long duration. One obvious advantage is crew safety, each spacecraft being designed to accept the crew of the sister ship in the event of a major failure. This approach also allows more exploration equipment to be carried on the expedition and enhances the probability of achieving mission objectives.

In the current concept, each vehicle assembled in earth orbit consists of three nuclear propulsion modules (Nuclear Shuttles) side-by-side, with the planetary spacecraft docked to the center module. Each spacecraft is nominally capable of sustaining a crew of six people for two years, or a crew of twelve for an extended period, in case of emergency. The earth orbit departure maneuver is initiated with the firing of the outer two propulsion modules as illustrated.

EARTH ORBIT DEPARTURE MANEUVERS [3]

The two planetary vehicles are assembled in a circular Earth orbit. Each vehicle consists of a spacecraft, two nuclear shuttle vehicles for Earth departure propulsion, and one nuclear shuttle for the remaining propulsion requirements through the Mars mission.

Following assembly and checkout in Earth orbit, each of the planetary ships is accelerated by the two outer nuclear shuttles to trans-Mars injection velocity. The two outer nuclear shuttles are then shut down, separated from the planetary vehicle, oriented for retro-fire, and then retro-fired to place them on a highly elliptic path returning to the original assembly orbit altitude. After a coast of several days, the nuclear shuttles arrive at the original assembly orbit altitude and are retro-fired again to place them into a circular orbit. The nuclear shuttles rendezvous with a space station to be checked out and refueled for further utilization. The nuclear shuttles which return to Earth orbit will be available for transfer of fuel and supplies to geosynchronous orbit or to lunar orbit.

MISSION WEIGHT HISTORY [4]

Each completely assembled and fueled vehicle for the 1981 Manned Mars Landing Mission weighs approximately 1.6 million pounds just prior to the Earth orbit departure maneuver. As the vehicle reaches the necessary earth departure velocity and the outer two Nuclear Shuttles are separated, its weight has decreased to 675,000 pounds. Thus nearly 60 of the initial vehicle weight is required for the Earth departure maneuver.

During the Earth-to-Mars phase of the mission, the only significant weight losses are due to midcourse corrections and life support consumables which amount to about 25,000 pounds. Thus at Mars arrival, just prior to the propulsive braking into Mars orbit, the vehicle weighs 650,000 pounds. While in Mars orbit, all the surface sample return probes and the Mars Excursion Module are deployed. These and other losses reduce the vehicle weight to 380,000 pounds just prior to Mars departure.

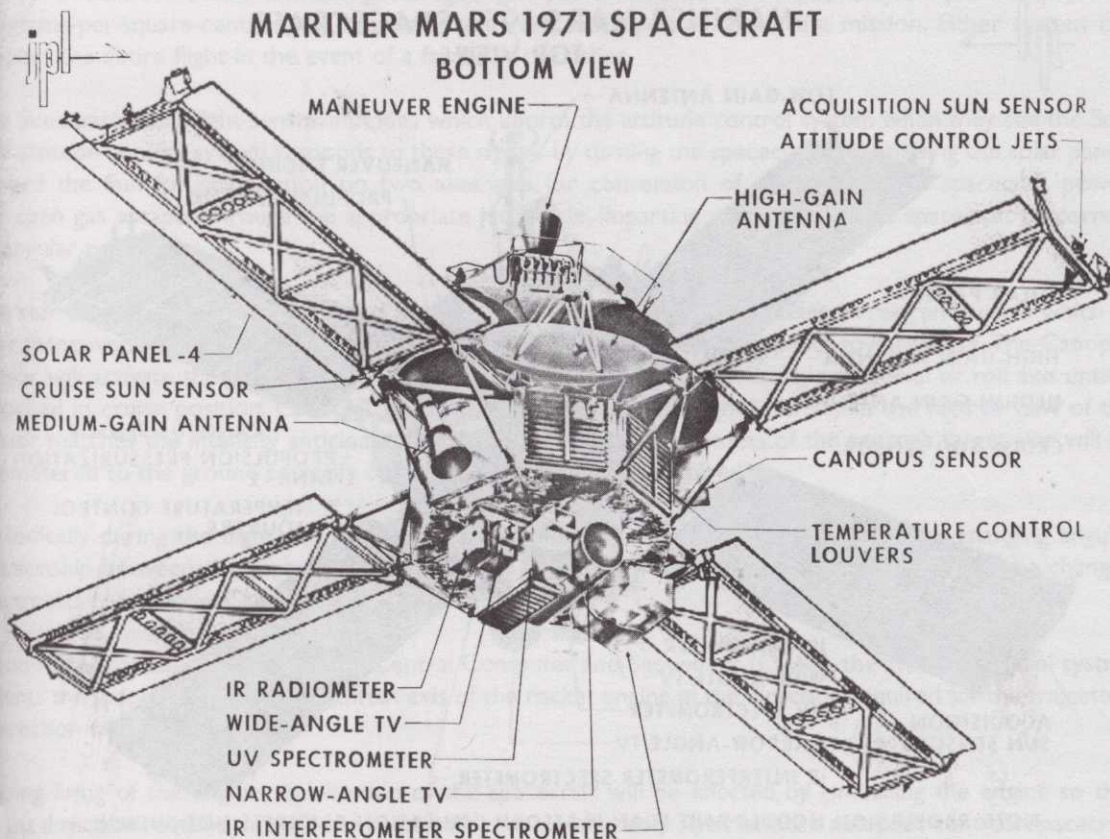
Midcourse corrections and life support consumables are again lost during the Mars-to-Earth phase of the trip. Also lost are the probes deployed at Venus encounter. As the vehicle approaches Earth for the final propulsive maneuver, it weighs 190,000 pounds. The total weight returned to earth orbit, consisting of the empty Nuclear Shuttle and the Mission Module (minus expendables) is about 160,000 pounds. Of the initial weight of 1.6 million pounds in Earth orbit, approximately 1.2 million, or nearly 75%, are propellant.

EN ROUTE SPACECRAFT CONFIGURATION [5]

The forward compartment of the spacecraft is an unpressurized area housing the Mars Excursion Module (MEM), an airlock to provide for pressurized transfer to the MEM and for extra vehicular activities, and unmanned probes. Six Mars Surface Sample Return probes and two Venus probes are carried on each spacecraft. Immediately aft of the airlock is the Mission Module which provides the crew with a shirt sleeve environment, living quarters, space vehicle control capability, experiment laboratories, radiation shelter, etc. The functional areas are distributed on four decks. This compartment is occupied by the crew of six for the entire mission except during the Mars surface activity.

MARINER MARS 1971 SPACECRAFT

BOTTOM VIEW



NOTE: PROPULSION MODULE AND SCAN PLATFORM INSULATION BLANKETS NOT SHOWN

between two solar panels. The cruise Sun sensor and Sun gate are attached to a solar panel outrigger.

Eight electronics compartments girdling the spacecraft house the following: Bay 1, power regulators; Bay 2, power conversion equipment, scan control and IRIS electronics; Bay 3, Central Computer and Sequencer; Bay 4, attitude control sub-system; Bay 5, flight telemetry and command subsystems; Bay 6, tape recorder; Bay 7, radio receiver and transmitters; Bay 8, television electronics and data automation subsystem; Bay 9, spacecraft battery.

Four of the electronics compartments are temperature controlled by lightweight louver assemblies on the outer surfaces. The octagon's interior is insulated by multilayer fabric thermal blankets at both top and bottom of the structure.

To avoid puncture of the propellant tanks by micrometeoroids, the outer layer of the top thermal blanket is constructed of a tightly-woven fiberglass cloth (Armalon) designed to break up striking particles.

The Mariners will carry science instrumentation for four planetary experiments. Two additional experiments — spacecraft occultation by Mars and celestial mechanics — require only the spacecraft communications system as the source of their data.

The television cameras, an infrared interferometer spectrometer (IRIS), ultraviolet spectrometer (UVS) and infrared radiometer (IRR) are mounted on a motor-driven two-degree-of-freedom scan platform on the bottom or shaded side of the octagon. Total rotating weight of the platform mechanism and its science instrument payload is 181 pounds (82.2 kilograms).

The Mariner weighs 1,200 pounds (544 kilograms) unfueled and measures 7½ feet (2.29 meters) from the scan platform to the top of the low-gain antenna and rocket nozzle. With solar panels deployed, the